

Correlation and Simple Linear Regression

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This page reviews two ways to relate two continuous variables: correlation coefficients, with tests of whether they differ from zero, and simple linear regression. It uses the t reference distribution defined on the [Statistical Inference](#) page. This page is adapted from Vittinghoff et al. (2012), Chapter 3.

The HERS data

The examples on this page use the HERS data, which the [Comparing Means](#) page describes. The `rmb` R package includes the dataset; `haven::as_factor()` converts its Stata value labels to factors:

```
hers <- rmb::hers |> haven::as_factor()
```

Correlation

Testing the Pearson correlation

Definition 0.1 (Population correlation). The **population correlation** of two random variables X and Y with positive variances is

$$\rho \stackrel{\text{def}}{=} \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X) \text{Var}(Y)}},$$

where $\text{Cov}(X, Y)$ is their [covariance](#).

The [Pearson correlation coefficient](#) r of a sample is the corresponding sample statistic, and estimates ρ .

Definition 0.2 (Test of zero correlation). Let r be the [Pearson correlation coefficient](#) of $n \geq 3$ pairs, with $|r| < 1$. The **t-test of zero correlation** of $H_0 : \rho = 0$ against $H_1 : \rho \neq 0$ (Definition [0.1](#)) uses the statistic

$$t \stackrel{\text{def}}{=} \frac{r\sqrt{n-2}}{\sqrt{1-r^2}}.$$

Its p-value is $\Pr(|T| \geq |t|)$, where T has the t_{n-2} distribution ([t-distribution](#)).

Theorem 0.1 (Null distribution of the correlation t statistic). *Let the pairs $(X_1, Y_1), \dots, (X_n, Y_n)$ be independent, and let each Y_i , given $X_1, \dots, X_n$, be Gaussian with a mean and variance that do not depend on the X values. Then the statistic t of Definition [0.2](#) has the t_{n-2} distribution (Hogg et al. 2019, sec. 9.6, pp. 472-473).*

The conditions of Theorem [0.1](#) hold, for example, when the pairs are independent draws from a bivariate Gaussian distribution with $\rho = 0$.

Example 0.1 (Correlation between BMI and fasting glucose in HERS). Figure [1](#) plots baseline fasting glucose against BMI.

```

hers |>
  dplyr::filter(!is.na(BMI)) |>
  ggplot2::ggplot() +
  ggplot2::aes(x = BMI, y = glucose) +
  ggplot2::geom_point(alpha = 0.3) +
  ggplot2::geom_smooth(method = "lm", formula = y ~ x) +
  ggplot2::labs(
    x = "BMI (kg/m^2)",
    y = "Fasting glucose (mg/dL)"
  )

```

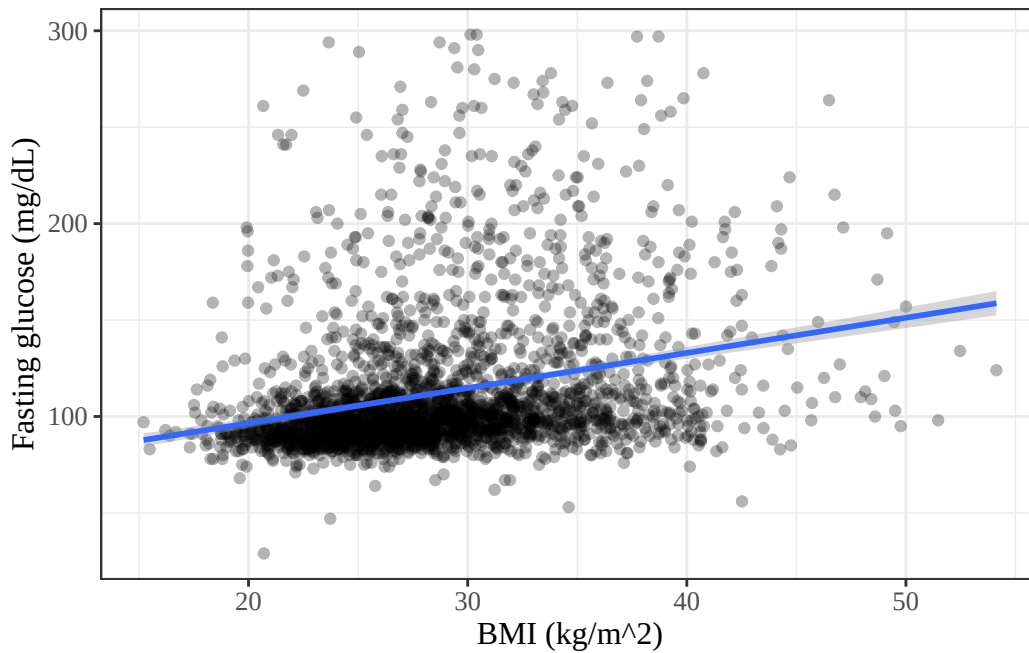


Figure 1: Baseline fasting glucose against BMI in HERS, with the least-squares line

The statistic of Definition 0.2, using the 2758 participants with a BMI measurement:

```

hers_bmi <- hers |> dplyr::filter(!is.na(BMI))
r <- cor(hers_bmi$BMI, hers_bmi$glucose)
n <- nrow(hers_bmi)
t_cor <- r * sqrt(n - 2) / sqrt(1 - r^2)
c(r = r, t = t_cor, p_value = 2 * pt(-abs(t_cor), df = n - 2))
#>           r           t      p_value
#> 2.72664e-01 1.48779e+01 3.24201e-48

```

`cor.test()` reports the same values, and a confidence interval for ρ :

```
cor.test(hers$BMI, hers$glucose, method = "pearson")
#>
#> Pearson's product-moment correlation
#>
#> data:  hers$BMI and hers$glucose
#> t = 14.88, df = 2756, p-value <2e-16
#> alternative hypothesis: true correlation is not equal to 0
#> 95 percent confidence interval:
#>  0.237760 0.306865
#> sample estimates:
#>      cor
#> 0.272664
```

The correlation is positive but modest: glucose tends to be higher at higher BMI, with wide scatter around the line in Figure 1.

Spearman rank correlation

Definition 0.3 (Spearman rank correlation). Replace each x_i by its rank among $x_1, \dots, x_n$, and each y_i by its rank among $y_1, \dots, y_n$, giving tied values the average of the ranks they span. The **Spearman rank correlation** r_S is the [Pearson correlation coefficient](#) of the ranks.

Because ranks depend only on the ordering of the values, r_S measures how close the association is to monotone, whether or not it is linear, and a single extreme value moves r_S less than it moves r .

Example 0.2 (Spearman correlation between BMI and fasting glucose in HERS). The Pearson correlation of the ranks (Definition 0.3), and `cor.test()`'s Spearman test (`exact = FALSE`, because tied values rule out the exact p-value):

```

hers_bmi <- hers |> dplyr::filter(!is.na(BMI))
cor(rank(hers_bmi$BMI), rank(hers_bmi$glucose))
#> [1] 0.333751
cor.test(hers_bmi$BMI, hers_bmi$glucose, method = "spearman", exact = FALSE)
#>
#> Spearman's rank correlation rho
#>
#> data:  hers_bmi$BMI and hers_bmi$glucose
#> S = 2.33e+09, p-value <2e-16
#> alternative hypothesis: true rho is not equal to 0
#> sample estimates:
#>      rho
#> 0.333751

```

r_S is larger than the Pearson r of Example 0.1: the association is closer to monotone than to linear.

Simple linear regression

Model specification

Definition 0.4 (Simple linear regression model). A **simple linear regression** model relates a continuous outcome Y to a single predictor X :

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \quad \varepsilon_1, \dots, \varepsilon_n \sim_{\text{iid}} N(0, \sigma^2).$$

- β_0 is the **intercept**: the mean of Y among observations with $X = 0$.
- β_1 is the **slope**: the difference in the mean of Y between two groups whose values of X differ by one unit, $\beta_1 = E[Y \mid X = x + 1] - E[Y \mid X = x]$.
- σ^2 is the variance of Y around its mean at each value of X .

Ordinary least squares estimation

Definition 0.5 (Residual sum of squares). For data $(x_1, y_1), \dots, (x_n, y_n)$, the **residual sum of squares** of a line with intercept b_0 and slope b_1 is

$$\text{RSS}(b_0, b_1) \stackrel{\text{def}}{=} \sum_{i=1}^n (y_i - b_0 - b_1 x_i)^2.$$

Example 0.3 (Residual sum of squares of a line through three points). For the points $(0, 1)$, $(1, 2)$, $(2, 2)$ and the line with $b_0 = 1$ and $b_1 = 0.5$, the vertical distances from the points to the line are $1 - 1 = 0$, $2 - 1.5 = 0.5$, and $2 - 2 = 0$, so $\text{RSS}(1, 0.5) = 0^2 + 0.5^2 + 0^2 = 0.25$.

Definition 0.6 (Ordinary least squares). The **ordinary least squares (OLS) estimates** $\hat{\beta}_0$ and $\hat{\beta}_1$ are the values of b_0 and b_1 that minimize the **residual sum of squares** $\text{RSS}(b_0, b_1)$.

Theorem 0.2 (Closed-form OLS estimates). Write

$$S_{xx} \stackrel{\text{def}}{=} \sum_{i=1}^n (x_i - \bar{x})^2, \quad S_{yy} \stackrel{\text{def}}{=} \sum_{i=1}^n (y_i - \bar{y})^2, \quad S_{xy} \stackrel{\text{def}}{=} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}),$$

and suppose $S_{xx} > 0$ (not all x_i are equal). Then the OLS estimates (Definition 0.6) are unique, and

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}}, \quad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}.$$

i Proof

Proof. RSS is a quadratic function of (b_0, b_1) , so we find where its partial derivatives are zero, then check that this point is the unique minimum.

The derivative with respect to b_0 :

$$\begin{aligned} \frac{\partial \text{RSS}}{\partial b_0} &= -2 \sum_{i=1}^n (y_i - b_0 - b_1 x_i) \quad (\text{chain rule, term by term}) \\ &= -2(n\bar{y} - nb_0 - b_1 n\bar{x}) \quad \left(\sum_i y_i = n\bar{y}, \sum_i x_i = n\bar{x} \right) \end{aligned}$$

Setting it to zero and dividing by $-2n$ gives $\bar{y} - b_0 - b_1 \bar{x} = 0$, so $b_0 = \bar{y} - b_1 \bar{x}$.

The derivative with respect to b_1 , after substituting $b_0 = \bar{y} - b_1 \bar{x}$:

$$\begin{aligned} \frac{\partial \text{RSS}}{\partial b_1} &= -2 \sum_{i=1}^n x_i (y_i - b_0 - b_1 x_i) \quad (\text{chain rule, term by term}) \\ &= -2 \sum_{i=1}^n x_i ((y_i - \bar{y}) - b_1 (x_i - \bar{x})) \quad (\text{substitute } b_0) \\ &= -2 \sum_{i=1}^n (x_i - \bar{x}) ((y_i - \bar{y}) - b_1 (x_i - \bar{x})) \quad \left(\text{subtract } \bar{x} \sum_i ((y_i - \bar{y}) - b_1 (x_i - \bar{x})) = 0 \right) \\ &= -2(S_{xy} - b_1 S_{xx}) \quad (\text{definitions of } S_{xy} \text{ and } S_{xx}) \end{aligned}$$

The subtracted sum in the third step is zero because $\sum_i (y_i - \bar{y}) = 0$ and $\sum_i (x_i - \bar{x}) = 0$.

Setting the derivative to zero gives $b_1 = S_{xy}/S_{xx}$.

This stationary point is the unique minimum. The matrix of second derivatives of RSS is

$$2 \begin{pmatrix} n & n\bar{x} \\ n\bar{x} & \sum_i x_i^2 \end{pmatrix},$$

whose top-left entry $2n$ is positive and whose determinant is $4(n \sum_i x_i^2 - n^2 \bar{x}^2) = 4nS_{xx} > 0$. So the matrix is positive definite, RSS is strictly convex, and its only stationary point is its global minimum. $\square$

Corollary 0.1 (OLS slope in terms of the correlation). *If $S_{xx} > 0$ and $S_{yy} > 0$, then*

$$\hat{\beta}_1 = r \frac{s_y}{s_x},$$

where r is the *Pearson correlation coefficient* and s_x and s_y are the *sample standard deviations* of the x_i and the y_i .

i Proof

Proof. With the notation of Theorem 0.2, $r = S_{xy}/\sqrt{S_{xx}S_{yy}}$, $s_x = \sqrt{S_{xx}/(n-1)}$, and $s_y = \sqrt{S_{yy}/(n-1)}$. So:

$$\begin{aligned} r \frac{s_y}{s_x} &= \frac{S_{xy}}{\sqrt{S_{xx}S_{yy}}} \cdot \frac{\sqrt{S_{yy}/(n-1)}}{\sqrt{S_{xx}/(n-1)}} && \text{(substitute } r, s_x, s_y) \\ &= \frac{S_{xy}}{\sqrt{S_{xx}}\sqrt{S_{yy}}} \cdot \frac{\sqrt{S_{yy}}}{\sqrt{S_{xx}}} && \text{(cancel the factors of } n-1) \\ &= \frac{S_{xy}}{S_{xx}} && \text{(cancel } \sqrt{S_{yy}}) \\ &= \hat{\beta}_1 && \text{(closed-form OLS slope)} \end{aligned}$$

$\square$

Fitting a simple linear regression in R

Example 0.4 (Regression of fasting glucose on BMI in HERS). The OLS estimates from Theorem 0.2, and the slope from Corollary 0.1, for the participants with a BMI measurement:

```

hers_bmi <- hers |> dplyr::filter(!is.na(BMI))
x <- hers_bmi$BMI
y <- hers_bmi$glucose
b1 <- sum((x - mean(x)) * (y - mean(y))) / sum((x - mean(x))^2)
c(
  intercept = mean(y) - b1 * mean(x),
  slope = b1,
  slope_from_r = cor(x, y) * sd(y) / sd(x)
)
#>      intercept      slope slope_from_r
#>      60.07405      1.82218      1.82218

```

`lm()` gives the same estimates:

```

slr_fit <- lm(glucose ~ BMI, data = hers)
summary(slr_fit)
#>
#> Call:
#> lm(formula = glucose ~ BMI, data = hers)
#>
#> Residuals:
#>      Min       1Q   Median       3Q      Max
#> -81.55 -18.98 -10.35   3.76 190.81
#>
#> Coefficients:
#>              Estimate Std. Error t value Pr(>|t|)
#> (Intercept)   60.074      3.565    16.9   <2e-16 ***
#> BMI           1.822       0.122    14.9   <2e-16 ***
#> ---
#> Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
#>
#> Residual standard error: 35.5 on 2756 degrees of freedom
#> (5 observations deleted due to missingness)
#> Multiple R-squared:  0.0743, Adjusted R-squared:  0.074
#> F-statistic: 221 on 1 and 2756 DF, p-value: <2e-16

```

The estimated slope is $\hat{\beta}_1 = 1.82$ mg/dL per kg/m²: mean fasting glucose is about 1.8 mg/dL higher among participants whose BMI is 1 kg/m² higher. The t statistic for the slope equals the correlation test statistic of Example 0.1.

The coefficient of determination

Definition 0.7 (Total sum of squares). The **total sum of squares** of $y_1, \dots, y_n$ is

$$\text{TSS} \stackrel{\text{def}}{=} \sum_{i=1}^n (y_i - \bar{y})^2.$$

Example 0.5 (Total sum of squares of three values). For $y = 1, 2, 2$, $\bar{y} = 5/3$, so

$$\begin{aligned} \text{TSS} &= \left(1 - \frac{5}{3}\right)^2 + 2\left(2 - \frac{5}{3}\right)^2 && \text{(definition)} \\ &= \frac{4}{9} + \frac{2}{9} && \text{(square the deviations)} \\ &= \frac{2}{3} && \text{(arithmetic)} \end{aligned}$$

Definition 0.8 (Coefficient of determination). For a fitted regression with fitted values $\hat{y}_i$, the **coefficient of determination** is

$$R^2 \stackrel{\text{def}}{=} 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}.$$

The numerator is the **residual sum of squares** of the fit, and the denominator is the **total sum of squares** of the y_i .

R^2 is often described as the proportion of the variation in Y explained by the regression on X .

Theorem 0.3 (R^2 of a simple linear regression). *For the OLS fit of a simple linear regression, with $S_{xx} > 0$ and $S_{yy} > 0$, $R^2 = r^2$, where r is the **Pearson correlation coefficient** of the x_i and y_i . In particular, $0 \leq R^2 \leq 1$.*

Proof

Proof. With the notation of Theorem 0.2, the fitted values are $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$, so each residual is

$$\begin{aligned} y_i - \hat{y}_i &= y_i - (\bar{y} - \hat{\beta}_1 \bar{x}) - \hat{\beta}_1 x_i && \text{(substitute } \hat{\beta}_0) \\ &= (y_i - \bar{y}) - \hat{\beta}_1 (x_i - \bar{x}) && \text{(regroup)} \end{aligned}$$

Squaring and summing:

$$\begin{aligned}
\sum_{i=1}^n (y_i - \hat{y}_i)^2 &= \sum_{i=1}^n ((y_i - \bar{y})^2 - 2\hat{\beta}_1(x_i - \bar{x})(y_i - \bar{y}) + \hat{\beta}_1^2(x_i - \bar{x})^2) && \text{(expand the square)} \\
&= S_{yy} - 2\hat{\beta}_1 S_{xy} + \hat{\beta}_1^2 S_{xx} && \text{(definitions of } S_{yy}, S_{xy}, S_{xx}) \\
&= S_{yy} - 2\frac{S_{xy}^2}{S_{xx}} + \frac{S_{xy}^2}{S_{xx}} && \text{(substitute } \hat{\beta}_1 = S_{xy}/S_{xx}) \\
&= S_{yy} - \frac{S_{xy}^2}{S_{xx}} && \text{(combine like terms)}
\end{aligned}$$

The total sum of squares is S_{yy} , so:

$$\begin{aligned}
R^2 &= 1 - \frac{S_{yy} - S_{xy}^2/S_{xx}}{S_{yy}} && \text{(substitute into the definition of } R^2) \\
&= \frac{S_{xy}^2}{S_{xx}S_{yy}} && \text{(simplify)} \\
&= r^2 && (r = S_{xy}/\sqrt{S_{xx}S_{yy}})
\end{aligned}$$

Since $-1 \leq r \leq 1$ ([range of the correlation coefficient](#)), $0 \leq r^2 \leq 1$. □

Example 0.6 (R^2 for the regression of glucose on BMI in HERS). For the fit in Example 0.4, R^2 computed from Definition 0.8, the square of the Pearson correlation, and `lm()`'s value agree:

```

fit <- lm(glucose ~ BMI, data = hers)
y <- fit$model$glucose
c(
  r_squared = 1 - sum(residuals(fit)^2) / sum((y - mean(y))^2),
  cor_squared = cor(fit$model$BMI, y)^2,
  lm = summary(fit)$r.squared
)
#>   r_squared cor_squared      lm
#> 0.0743455 0.0743455 0.0743455

```

BMI accounts for only about 7% of the variation in baseline fasting glucose.

Further reading

[Linear Models Overview](#) covers linear regression in depth, including inference for the coefficients and multiple predictors. Vittinghoff et al. (2012) cover linear regression in Chapter 4.

References

- Hogg, Robert V., Elliot A. Tanis, and Dale L. Zimmerman. 2019. *Probability and Statistical Inference*. Tenth edition. Pearson.
- Vittinghoff, Eric, David V Glidden, Stephen C Shiboski, and Charles E McCulloch. 2012. *Regression Methods in Biostatistics: Linear, Logistic, Survival, and Repeated Measures Models*. 2nd ed. Springer. <https://doi.org/10.1007/978-1-4614-1353-0>.